

# Enhancing Next Activity Prediction in Process Mining with Retrieval-Augmented Generation

Appendix: complete per-log evaluation results

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## Setup

This appendix reports one table per event log, with all ten evaluated LLMs and all metrics. Every table reports accuracy, macro-averaged precision, recall and F1 score, the total execution time in hours and the average per-inference time in seconds, for the ten benchmarked models: Llama-3.1-8B-Instruct, Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, Mistral-7B-Instruct-v0.2, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct, Phi-4, DeepSeek-R1-Distill-Qwen-7B, DeepSeek-R1-Distill-Llama-8B and gpt-4o-mini.

Results were obtained on a workstation running Ubuntu 22.04.3 LTS with an NVIDIA A100 GPU, using the single-bucket strategy and the all-MiniLM-L12-v2 embedding model. Tables 1–7 cover the seven real-world logs and Tables 8–9 the two synthetic ones.

**Reading the tables.** Best value per column in **bold**, second best underlined, as in the published article these results come from.

## Real-world event logs

**Table 1.** Next activity prediction on the *BPIC 2012* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.53        | 0.43        | 0.38        | 0.39        | 4.15        | 49.8         |
| Llama-3.2-1B-Instruct        | 0.17        | 0.33        | 0.19        | 0.21        | <b>0.85</b> | <b>10.3</b>  |
| Llama-3.2-3B-Instruct        | 0.27        | 0.24        | 0.20        | 0.20        | 3.44        | 41.3         |
| Mistral-7B-Instruct-v0.2     | 0.42        | 0.42        | 0.37        | 0.38        | 2.76        | 33.2         |
| Mistral-7B-Instruct-v0.3     | 0.46        | 0.51        | 0.48        | 0.47        | 1.92        | <u>19.4</u>  |
| Qwen2.5-7B-Instruct          | 0.57        | 0.52        | 0.51        | 0.51        | 3.43        | 41.1         |
| Phi-4                        | <u>0.67</u> | <u>0.63</u> | <u>0.67</u> | <u>0.63</u> | 3.39        | 40.7         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.47        | 0.45        | 0.44        | 0.44        | 4.21        | 50.5         |
| DeepSeek-R1-Distill-Llama-8B | 0.55        | 0.55        | 0.53        | 0.54        | 7.78        | 93.3         |
| gpt-4o-mini                  | <b>0.76</b> | <b>0.72</b> | <b>0.68</b> | <b>0.67</b> | <u>1.68</u> | 20.1         |

**Table 2.** Next activity prediction on the *BPIC 2013 CP* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.72        | 0.70        | 0.71        | 0.70        | 3.54        | 42.4         |
| Llama-3.2-1B-Instruct        | 0.17        | 0.17        | 0.16        | 0.16        | 1.21        | 14.5         |
| Llama-3.2-3B-Instruct        | 0.28        | 0.26        | 0.27        | 0.26        | 1.96        | 23.5         |
| Mistral-7B-Instruct-v0.2     | 0.42        | 0.37        | 0.37        | 0.36        | 2.98        | 35.7         |
| Mistral-7B-Instruct-v0.3     | 0.66        | 0.55        | 0.57        | 0.55        | <b>0.52</b> | <b>6.25</b>  |
| Qwen2.5-7B-Instruct          | 0.83        | 0.78        | <u>0.79</u> | 0.78        | 3.52        | 42.1         |
| Phi-4                        | <u>0.89</u> | <u>0.85</u> | <b>0.87</b> | <u>0.86</u> | 5.96        | 71.5         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.62        | 0.58        | 0.57        | 0.57        | 7.43        | 89.16        |
| DeepSeek-R1-Distill-Llama-8B | 0.67        | 0.63        | 0.62        | 0.63        | 14.68       | 176.16       |
| gpt-4o-mini                  | <b>0.90</b> | <b>0.87</b> | <b>0.87</b> | <b>0.87</b> | <u>1.13</u> | <u>13.6</u>  |

**Table 3.** Next activity prediction on the *BPIC 2013 I* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.37        | 0.33        | 0.32        | 0.32        | 5.28        | 63.3         |
| Llama-3.2-1B-Instruct        | 0.03        | 0.03        | 0.02        | 0.03        | <u>2.65</u> | <u>31.8</u>  |
| Llama-3.2-3B-Instruct        | 0.05        | 0.06        | 0.05        | 0.05        | 3.05        | 36.5         |
| Mistral-7B-Instruct-v0.2     | 0.38        | 0.31        | 0.30        | 0.30        | 4.49        | 53.8         |
| Mistral-7B-Instruct-v0.3     | 0.12        | 0.12        | 0.11        | 0.11        | 4.38        | 52.5         |
| Qwen2.5-7B-Instruct          | 0.45        | 0.32        | 0.32        | 0.31        | 2.71        | 32.5         |
| Phi-4                        | <u>0.53</u> | <u>0.57</u> | <u>0.53</u> | <u>0.54</u> | 5.15        | 61.7         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.39        | 0.38        | 0.37        | 0.37        | 5.69        | 68.2         |
| DeepSeek-R1-Distill-Llama-8B | 0.46        | 0.43        | 0.42        | 0.43        | 9.52        | 114.2        |
| gpt-4o-mini                  | <b>0.70</b> | <b>0.64</b> | <b>0.65</b> | <b>0.64</b> | <b>1.79</b> | <b>21.5</b>  |

**Table 4.** Next activity prediction on the *BPIC 2020 ID* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.71        | 0.74        | 0.68        | 0.67        | 2.65        | 31.8         |
| Llama-3.2-1B-Instruct        | 0.23        | 0.31        | 0.18        | 0.19        | <b>0.71</b> | <b>8.5</b>   |
| Llama-3.2-3B-Instruct        | 0.39        | 0.61        | 0.42        | 0.48        | 2.05        | 24.5         |
| Mistral-7B-Instruct-v0.2     | 0.79        | 0.74        | 0.71        | 0.71        | 2.34        | 28.1         |
| Mistral-7B-Instruct-v0.3     | 0.63        | 0.50        | 0.47        | 0.46        | 1.29        | 15.4         |
| Qwen2.5-7B-Instruct          | 0.75        | 0.82        | 0.80        | 0.78        | 2.13        | 25.5         |
| Phi-4                        | <u>0.93</u> | <u>0.94</u> | <u>0.93</u> | <u>0.93</u> | 3.12        | 37.3         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.69        | 0.59        | 0.58        | 0.55        | 3.93        | 47.2         |
| DeepSeek-R1-Distill-Llama-8B | 0.70        | 0.70        | 0.69        | 0.67        | 7.34        | 88.1         |
| gpt-4o-mini                  | <b>0.98</b> | <b>0.99</b> | <b>0.99</b> | <b>0.99</b> | <u>0.92</u> | <u>11.0</u>  |

**Table 5.** Next activity prediction on the *Hospital Billing* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.55        | 0.64        | 0.63        | 0.59        | 2.73        | 32.7         |
| Llama-3.2-1B-Instruct        | 0.12        | 0.31        | 0.09        | 0.12        | <b>0.86</b> | <b>10.3</b>  |
| Llama-3.2-3B-Instruct        | 0.15        | 0.31        | 0.16        | 0.19        | 2.26        | 27.1         |
| Mistral-7B-Instruct-v0.2     | 0.51        | 0.55        | 0.48        | 0.50        | 3.16        | 37.9         |
| Mistral-7B-Instruct-v0.3     | 0.60        | 0.63        | 0.61        | 0.61        | 2.28        | 27.4         |
| Qwen2.5-7B-Instruct          | 0.69        | <b>0.72</b> | <u>0.74</u> | <b>0.72</b> | 2.31        | 27.6         |
| Phi-4                        | <u>0.71</u> | 0.70        | <b>0.78</b> | <u>0.71</u> | 3.85        | 46.1         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.63        | 0.68        | 0.68        | 0.67        | 4.58        | 54.9         |
| DeepSeek-R1-Distill-Llama-8B | 0.56        | 0.55        | 0.58        | 0.53        | 9.58        | 114.9        |
| gpt-4o-mini                  | <b>0.73</b> | <u>0.71</u> | <u>0.74</u> | <u>0.71</u> | <u>1.19</u> | <u>14.2</u>  |

**Table 6.** Next activity prediction on the *Sepsis Cases* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.53        | 0.60        | 0.60        | 0.58        | 2.98        | 35.7         |
| Llama-3.2-1B-Instruct        | 0.18        | 0.39        | 0.23        | 0.24        | 1.05        | 12.5         |
| Llama-3.2-3B-Instruct        | 0.26        | 0.41        | 0.27        | 0.31        | 2.41        | 28.8         |
| Mistral-7B-Instruct-v0.2     | 0.50        | 0.66        | 0.59        | 0.61        | 2.74        | 32.8         |
| Mistral-7B-Instruct-v0.3     | 0.55        | 0.67        | 0.68        | 0.64        | <b>0.71</b> | <b>8.54</b>  |
| Qwen2.5-7B-Instruct          | 0.66        | <u>0.80</u> | <u>0.79</u> | <u>0.79</u> | 2.87        | 34.4         |
| Phi-4                        | <b>0.70</b> | 0.77        | 0.76        | 0.77        | 3.47        | 41.6         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.55        | 0.56        | 0.60        | 0.54        | 6.86        | 82.3         |
| DeepSeek-R1-Distill-Llama-8B | 0.54        | 0.54        | 0.61        | 0.55        | 13.14       | 157.6        |
| gpt-4o-mini                  | <u>0.69</u> | <b>0.85</b> | <b>0.83</b> | <b>0.84</b> | <u>0.90</u> | <u>10.8</u>  |

**Table 7.** Next activity prediction on the *Udon-ya* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.56        | 0.57        | 0.51        | 0.53        | 3.00        | 36.0         |
| Llama-3.2-1B-Instruct        | 0.14        | 0.27        | 0.11        | 0.11        | <b>0.71</b> | <b>8.4</b>   |
| Llama-3.2-3B-Instruct        | 0.38        | 0.48        | 0.30        | 0.36        | 2.36        | 28.3         |
| Mistral-7B-Instruct-v0.2     | 0.54        | 0.54        | 0.45        | 0.47        | 2.39        | 28.6         |
| Mistral-7B-Instruct-v0.3     | 0.63        | 0.54        | 0.50        | 0.50        | <u>1.16</u> | <u>13.92</u> |
| Qwen2.5-7B-Instruct          | <u>0.72</u> | <u>0.75</u> | <u>0.68</u> | <u>0.70</u> | 2.46        | 29.5         |
| Phi-4                        | 0.65        | 0.69        | 0.61        | 0.63        | 3.71        | 44.5         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.61        | 0.49        | 0.48        | 0.47        | 3.73        | 44.7         |
| DeepSeek-R1-Distill-Llama-8B | 0.58        | 0.55        | 0.50        | 0.51        | 6.89        | 82.6         |
| gpt-4o-mini                  | <b>0.84</b> | <b>0.80</b> | <b>0.79</b> | <b>0.78</b> | 1.37        | 16.4         |

## Synthetic event logs

**Table 8.** Next activity prediction on the *SL2-2v-1r* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.39        | <u>0.17</u> | <u>0.16</u> | <u>0.16</u> | 0.33        | 25.8         |
| Llama-3.2-1B-Instruct        | 0.09        | 0.14        | 0.02        | 0.03        | <u>0.17</u> | <u>13.3</u>  |
| Llama-3.2-3B-Instruct        | 0.13        | 0.14        | 0.06        | 0.08        | 0.24        | 18.8         |
| Mistral-7B-Instruct-v0.2     | 0.18        | 0.10        | 0.09        | 0.08        | 0.50        | 39.1         |
| Mistral-7B-Instruct-v0.3     | 0.20        | 0.16        | 0.11        | 0.11        | 0.33        | 28.8         |
| Qwen2.5-7B-Instruct          | <b>0.81</b> | <b>0.25</b> | <b>0.22</b> | <b>0.22</b> | 0.39        | 30.5         |
| Phi-4                        | <u>0.72</u> | 0.14        | 0.14        | 0.13        | 0.67        | 52.43        |
| DeepSeek-R1-Distill-Qwen-7B  | 0.69        | 0.14        | 0.13        | 0.13        | 0.50        | 39.1         |
| DeepSeek-R1-Distill-Llama-8B | 0.58        | 0.13        | 0.10        | 0.11        | 1.34        | 104.8        |
| gpt-4o-mini                  | 0.70        | 0.14        | 0.14        | 0.14        | <b>0.13</b> | <b>10.4</b>  |

**Table 9.** Next activity prediction on the *SL3-MV* log.

| LLM                          | Acc.        | Prec.       | Rec.        | F1          | Time        |              |
|------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              |             |             |             |             | Total (h)   | Per-Inf. (s) |
| Llama-3.1-8B-Instruct        | 0.25        | 0.10        | 0.09        | 0.09        | 2.40        | 28.8         |
| Llama-3.2-1B-Instruct        | 0.07        | 0.07        | 0.01        | 0.02        | <b>0.82</b> | <b>9.8</b>   |
| Llama-3.2-3B-Instruct        | 0.42        | 0.06        | 0.06        | 0.06        | 1.89        | 22.6         |
| Mistral-7B-Instruct-v0.2     | 0.23        | 0.11        | 0.07        | 0.08        | 3.44        | 41.2         |
| Mistral-7B-Instruct-v0.3     | 0.22        | 0.11        | 0.08        | 0.09        | 2.82        | 33.8         |
| Qwen2.5-7B-Instruct          | <u>0.46</u> | <b>0.14</b> | <b>0.14</b> | <b>0.14</b> | 2.44        | 29.2         |
| Phi-4                        | 0.42        | 0.12        | 0.10        | 0.11        | 3.83        | 46.0         |
| DeepSeek-R1-Distill-Qwen-7B  | 0.42        | 0.12        | 0.10        | 0.11        | 2.94        | 35.3         |
| DeepSeek-R1-Distill-Llama-8B | 0.40        | 0.08        | 0.06        | 0.06        | 10.59       | 127.1        |
| gpt-4o-mini                  | <b>0.47</b> | <u>0.13</u> | <u>0.13</u> | <u>0.13</u> | <u>1.04</u> | <u>12.4</u>  |